

Problem

Find the inverse Laplace transform of:

(a) $H(s) = \frac{s+4}{s(s+2)}$

(b) $G(s) = \frac{s^2+4s+5}{(s+3)(s^2+2s+2)}$

(c) $F(s) = \frac{e^{-4s}}{s+2}$

(d) $D(s) = \frac{10s}{(s^2+1)(s^2+4)}$

Step-by-step solution

Step 1 of 4

(a). $H(s) = \frac{s+4}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2}$

$$s+4 = A(s+2) + Bs$$

Equating coefficients,

$$s: \quad 1 = A + B$$

$$\rightarrow \text{constant: } 4 = 2A \quad A=2, B=1-A=-1$$

$$H(s) = \frac{2}{s} - \frac{1}{s+2}$$

$$h(t) = 2u(t) - e^{-2t}u(t) = \underline{(2 - e^{-2t})u(t)}$$

Step 2 of 4

(b) $G(s) = \frac{A}{s+3} + \frac{Bs+C}{s^2+2s+2}$

$$s^2+4s+5 = (Bs+C)(s+3) + A(s^2+2s+2)$$

Equating coefficients,

$$s^2: \quad 1 = B + A \quad (1)$$

$$s: \quad 4 = 3B + C + 2A \quad (2)$$

$$\text{Constant: } 5 = 3C + 2A \quad (3)$$

Solving (1) to (3) gives

$$A = \frac{2}{5}, \quad B = \frac{3}{5}, \quad C = \frac{7}{5}$$

$$G(s) = \frac{0.4}{s+3} + \frac{0.6s+1.4}{s^2+2s+2} = \frac{0.4}{s+3} + \frac{0.6(s+1)+0.8}{(s+1)^2+1}$$

$$g(t) = \underline{0.4e^{-3t} + 0.6e^{-t} \cos t + 0.8e^{-t} \sin t} u(t)$$

Step 3 of 4

(c) $f(t) = \underline{e^{-2(t-4)}u(t-4)}$

Step 4 of 4

(d) $D(s) = \frac{10s}{(s^2+1)(s^2+4)} = \frac{As+B}{s^2+1} + \frac{Cs+D}{s^2+4}$

$$10s = (s^2+4)(As+B) + (s^2+1)(Cs+D)$$

Equating coefficients,

s³: $0 = A + C$

s²: $0 = B + D$

s: $10 = 4A + C$

constant: $0 = 4B + D$

Solving these leads to

$A = -10/3, B = 0, C = -10/3, D = 0$

$$D(s) = \frac{10s/3}{s^2+1} - \frac{10s/3}{s^2+4}$$

$$d(t) = \underline{\frac{10}{3}\cos t - \frac{10}{3}\cos 2t u(t)}$$